Finite-Deletion Bubble Certificates and Infinite-Operator Bounds for Percolation Nonuniqueness

A scoped analysis of the nonamenable quasi-transitive manuscript

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Abstract

The assigned September 24 manuscript claims the strict operator-threshold gap and a simultaneous nonuniqueness interval for every infinite nonamenable quasi-transitive graph. We map its probabilistic proof but do not certify that global theorem. Our main local deduction uses the actual closed-neighborhood deletion majorant for its corridor deformation: square summability of only the finitely many affected connection columns makes the deletion loss Hilbert–Schmidt. This remains meaningful when susceptibility is infinite and does not assume positivity of the deformed kernel. A finite Gram certificate has a quadratic discarded-column energy cost, while approximation of the deletion operator itself has a different, generally linear, cross-block cost. We prove an endpoint extension under almost-sure finite critical clusters, a uniform-tail criterion, and explicit regular-tree formulas. A separate whole-operator Schur and Neumann interface controls sprinkling and the already-present sandwich norm; finite vertex spectra alone supply none of these upper bounds. Classical operator tools are credited and the deep amplification, contact, escape, and phase arguments remain imported. Contribution assessment: C3/5, the source-specific compact deletion theorem and quantitative bubble-Gram certificates are notable local refinements, not an independent proof of the global nonuniqueness claim.

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1 Scope, status, and exact conventions

Let $G = (V, E)$ be a countably infinite, connected, locally finite, undirected graph with maximum degree $d < \infty$. The source's main setting further assumes quasi-transitivity and nonamenability. Bernoulli bond percolation has parameter p, and

$$T_p(x, y) = \mathbb{P}_p(x \leftrightarrow y), \qquad \chi_p(x) = \sum_y T_p(x, y).$$

All operator norms below are on *counting-measure* $\ell^2(V)$, unless explicitly stated. A kernel initially acts on finitely supported vectors; bounded extension is a conclusion or a hypothesis, never an automatic consequence of its finite entries. The thresholds are

$$p_c = \inf\{p : \mathbb{P}_p(\text{an infinite cluster exists}) = 1\}, \qquad p_{2\to 2} = \sup\{p : \|T_p\| < \infty\},$$

with p_u the uniqueness threshold. On the source's graph class, its claimed conclusion is $p_c < p_{2\to 2} \le p_u$. This report neither assumes this conclusion in its principal local theorem nor obtains it from finite computations.

Actual percolation kernels are symmetric and positive semidefinite: for finitely supported f,

$$\langle f, T_p f\rangle = \mathbb{E}_p \sum_{\text{clusters } C} \left|\sum_{x\in C} f(x)\right|^2 \ge 0.$$

This fact does *not* extend automatically to a kernel obtained by penalizing pivotal corridors. Entrywise inequalities and Loewner inequalities will therefore be distinguished throughout. For a finite set $L \subset V$, Π_L is the orthogonal coordinate projection. For a finite deleted set A, put

$$S = \mathrm{N}[A] = A \cup \{z : z \sim a \text{ for some } a \in A\}.$$

The neighbor collar is part of the source mechanism, not an optional enlargement that can be removed for a better constant.

1.1 Public literature and what was actually checked

Hutchcroft's published L^2-boundedness paper [2] formulates the general gap as a conjecture and develops its consequences. Its Remark 2.3 explicitly gives

$$T_p(x,y) \le 2\,\|T_p\| \exp\left(-\frac{\operatorname{dist}(x,y)}{e\,\|T_p\|}\right), \qquad 0 < p < p_{2\to 2}.$$

This is a consequence of a *known infinite operator bound*; it cannot be reversed by summing over an exponentially growing sphere without a sufficient decay rate. The earlier hyperbolic result [3] concerns an additional geometric hypothesis. Choi–Seo's 2025 preprint [4], whose abstract and metadata were freshly checked, claims a nonuniqueness phase for every Cayley graph of an acylindrically hyperbolic group. It is not a claim about every nonamenable quasi-transitive graph. Hutchcroft–Pan's Theorem 1.2 [5], freshly inspected in the primary HTML, states $p_{2\to 2} = p_u$ for the regular-tree products with infinite amenable Cayley graphs, and for the lamplighter on the tree. This equality is compatible with a strict gap above p_c.

The assigned publisher source [1] asserts the general result. Repository descriptions, including a formalization scope page, are not an independently replayed proof in this report. The literature refresh on October 9 used only public titles and URLs. Full external proofs were not all read, and no exhaustive novelty claim is made. The complete cached scientific manuscript, its figures' TeX, bibliography, and README were literally read in bounded displays.

2 The source proof and its indispensable obligations

The source's main achievement, if all its arguments hold, is not a finite spectral calculation. Its architecture couples the following quantitatively delicate modules.

Module	Role and retained obligation
Preliminaries	Simple-graph reduction, counting-measure operators, weighted mass transport, invariant trace, BK and Russo formulas, and the whole-operator sprinkling series.
Goodness	Capped critical clusters have controlled pivotal diameter after deletion. Centered exploration scores, marking, and injective inverse codes control exceptional events.
Amplification	A divergent critical moment yields finite-radius endpoint maps with success u, cap M, and conditional kernel K. Splicing and a joint inverse control second moments, eventually making $(\log(1/u)+\log M)/\log(1/\|K\|)$ small.
Moments	Domain potentials and size bias turn dispersive maps into susceptibility growth. A critical likelihood comparison and OSSS yield all critical moments of order below $1/2$, and almost mean-field subcritical susceptibility.
Trace	Two centered exploration scores bound large contacts between distinct clusters. A differentiated triangle trace, splitting at a cluster-size cutoff, yields a subpower bound on $\operatorname{tr}(T_p^3)$.
Deformation	Full-graph corridor weights calibrate a penalty, with small operator-norm loss but a quadratic decrease in averaged susceptibility once both exterior endpoint potentials are large.

Module	Role and retained obligation
Paths	Anchored geodesic pieces, a conditioned lazy middle walk, mass transport, collision bounds, and unconditioned smoothing supply two endpoint escapes with positive path mass.
Conclusion	Fixed geometric constants precede the choice $k = o(\log(1/(p_c - p)))$. Norm survival contradicts integrated row decay. Sprinkling and a compact-interval coupling give the phase statement simultaneously, not by an uncountable intersection.

Here tr denotes the source's finite invariant trace, obtained by averaging orbit representatives with normalized inverse-stabilizer weights in its unimodular reduction. Those weights do not change the Hilbert measure. Nonunimodular cases are imported from earlier literature. The symmetric lazy walk is

$$P(x,y) = \frac{1}{2d} \quad (x \sim y), \qquad P(x,x) = 1 - \frac{\deg x}{2d}, \qquad \|P\| = \rho < 1.$$

Its positivity and nonamenability estimate are genuine infinite-graph facts.

Three correct details are especially important. The splicing proof uses injective joint restoration for disjoint hanging pieces, not unjustified independence of the outputs. The contact proof compares two centered exploration scores without claiming they are independent. The path proof applies walk smoothing before imposing avoidance; conditioning first would invalidate that expectation estimate. Literal reading is not an independent certification of these deep arguments or of all the theorems they cite.

To locate the final quantitative interface, write

$$\delta = p_c - p, \quad m_0 = \|T_p\|, \quad R = \chi_p^+/m_0, \quad a_3 = \operatorname{tr}(T_p^3).$$

The source contradiction assumes unboundedness at criticality, gets $m_0 \geq (1-p)/(d\delta)$ by sprinkling, and imports $\log R, \log a_3 = o(\log(1/\delta))$. It chooses $k = O(\log(2R) + \log(2 + a_3) + \log\log(1/\delta))$, and a deformation time $t_* = B_*/m_0$, with B_* a fixed polynomial in kR. All its errors, including $t_* k c_k$ and the contact trace error, must be controlled with that same order of choices. Our finite-set results below do not supply these moving-parameter estimates.

3 The actual corridor deletion estimate

Fix the source's template length $k \geq 3$, and a base parameter $0 < p < 1$ in its stated subcritical regime. A raw P-path w has probability $\mu(w)$, loop-erased retained path γ_w of length $\ell_w \geq 3$, internal vertices I_w, and b_w blocker edges incident to I_w but not in γ_w. Its coefficient is

$$c_p(w) = \mu(w) p^{-\ell_w} (1-p)^{-b_w}.$$

It is a *full-graph* coefficient, unchanged when a domain D is used. Eligibility requires $\gamma_w \subset D$, not that every raw-walk vertex stays in D; absent blockers count as closed. For a connected pair x, y, N_{xy}^D sums these coefficients over counted open pivotal templates with closed blockers. Set

$$W_t^D(x,y) = \mathbf{1}_{\{x \leftrightarrow y \text{ in } D\}} e^{-tN_{xy}^D}, \qquad K_t^D(x,y) = \mathbb{E}_p W_t^D(x,y), \quad t \geq 0,$$

and set these kernels to zero at terminals outside D. The source proves domain monotonicity, symmetry, and $0 \leq K_t^D \leq T_p$ entrywise. No assertion $K_t^D \succeq 0$ is needed.

Lemma 3.1 (Source majorant, with its collar retained)**.** *For finite $A \subset D$ let $\Delta_t^A = K_t^D - K_t^{D\setminus A}$. Then*

$$0 \le \Delta_t^A(x,y) \le B_S(x,y) := \sum_{s\in S} T_p(x,s)T_p(s,y) \quad \textit{entrywise.} \tag{1}$$

Proof. We restate the source's local proof because it identifies the right finite interface. If deletion destroys an old connection, an old simple connection visits A. Otherwise a larger penalty can only come from a newly counted template. A previously nonpivotal edge becomes pivotal only if an old bypass visited A. A previously open blocker that is now absent joins A to an internal retained vertex outside A; that attachment lies on a surviving connection and belongs to S. Thus the entire positive loss, at most one, is bounded by an old simple x–y connection visiting S. Split that path at the visited s, use disjoint witnesses, BK, and a union bound. Domain monotonicity gives the lower inequality. There is no additional sum of template weights in this estimate. □

Summing (1) over y, the source obtains

$$\sum_y \Delta_t^A(x,y) \le \chi_p^+ \sum_{s\in S} T_p(x,s).$$

This is useful below p_c, but loses meaning when $\chi_p^+ = \infty$. Our change is to preserve the finite interface and measure its *column energy*, rather than summing the second leg in ℓ^1.

4 Finite-interface bubble and compactness theorem

For the next theorem T is any symmetric nonnegative kernel on a countable set, and Δ is a symmetric kernel obeying (1) with T in place of T_p. No boundedness of the whole operator T is assumed. Suppose

$$b_s := \sum_x T(x,s)^2 < \infty \quad (s \in S). \tag{2}$$

Define the finite-column map and its Gram matrix by

$$H : \mathbb{C}^S \longrightarrow \ell^2(V), \quad Hc = \sum_{s\in S} c_s T(\,\cdot\,,s), \qquad G_S = H^*H, \quad G_S(s,r) = \sum_x T(x,s)T(x,r).$$

These definitions make sense under (2), even if writing T^2 as a bounded operator would not make sense.

Theorem 4.1 (Compact deletion under local bubble control)**.** *Under (1)–(2), the deletion kernel extends to a self-adjoint Hilbert–Schmidt operator, and*

$$\|\Delta\| \le \lambda_{\max}(G_S), \qquad \|\Delta\|_{\mathrm{HS}} \le \left(\operatorname{tr} G_S^2\right)^{1/2}, \tag{3}$$

$$|\,\langle f, \Delta g\rangle\,| \le \||H^*|f|\|_{\mathbb{C}^S}\,\||H^*|g|\|_{\mathbb{C}^S} \quad (f,g \in \ell^2(V)). \tag{4}$$

In particular every Δ_t^A in Lemma 3.1 satisfies these estimates whenever its finitely many bubbles are finite. The same bounds work for every $t \ge 0$.

Proof. The majorant is the kernel of HH^*, a positive finite-rank operator. For finitely supported f, g, entrywise domination gives

$$|\,\langle f, \Delta g\rangle\,| \le \sum_{s\in S} \left(\sum_x |f(x)|T(x,s)\right)\left(\sum_y |g(y)|T(y,s)\right).$$

Cauchy–Schwarz in $\mathbb{C}^S$ proves (4), then density gives bounded extension with norm at most $\|H\|^2 = \|G_S\|$. Separately, Tonelli gives

$$\sum_{x,y} |\Delta(x,y)|^2 \le \sum_{x,y} B_S(x,y)^2 = \sum_{s,r\in S} G_S(s,r)^2 = \operatorname{tr} G_S^2 < \infty.$$

This proves the Hilbert–Schmidt assertion and compactness. Symmetry gives self-adjointness, but does not imply positivity. □

Remark 4.2 (Precisely what is new here). Finite-rank Gram factorization and Hilbert–Schmidt domination are classical tools. The source-specific deduction is that its collar-sensitive pivotal-template loss has this finite-interface structure, so a local bubble hypothesis replaces susceptibility finiteness for this task. Neither T nor K_t^D becomes compact. Even Δ need not have rank at most $|S|$; entrywise domination does not imply a rank bound.

For example, on $\mathbb{N}$, put $h_n = 2^{-n}$ and $\Delta_{ij} = h_i^2 \mathbf{1}_{\{i=j\}}$. Then $0 \le \Delta_{ij} \le h_i h_j$, but Δ has infinite rank. Even in dimension two, the matrix $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ is entrywise bounded by the rank-one all-ones matrix and has a negative eigenvalue. These examples explain why the proof did not use Loewner domination or a low-rank assertion for the actual loss.

4.1 A critical endpoint without a global norm assumption

Corollary 4.3 (Endpoint extension). *Suppose $p_* \in (0,1)$, all clusters at p_* are almost surely finite, and the source deletion comparison is available for parameters $p_n \uparrow p_*$. Fix k, t, D, A, and use the same raw walk templates and full-graph coefficient convention at each parameter. Then (1) passes to p_*. If the columns $T_{p_*}(\cdot, s)$, $s \in S$, are square summable, Theorem 4.1 applies at p_*, without assuming $\|T_{p_*}\| < \infty$ or $\chi_{p_*}^+ < \infty$.*

Proof. Couple edges by uniform labels. For a fixed terminal, its finite p_*-cluster and all its incident edges form a finite set. Almost surely no label in that set equals p_*; hence that cluster and all its internal bits agree with the p_n-configuration for all large n. This applies simultaneously to the two terminals and to induced-domain clusters. The retained part of a counted template lies in the relevant finite cluster. Its raw path may leave that cluster, but has fixed length k, so it stays in its finite k-neighborhood; only finitely many raw candidates exist. Their weights vary continuously with p. Thus the two W's converge almost surely and are bounded by one. Bounded convergence applies to their expectations. The right side of (1) has only finitely many summands; ordinary connection probabilities converge under this coupling. Apply the theorem. □

For the source's critical application, critical cluster finiteness is an imported probabilistic result, not established by a Gram calculation. This extension keeps k, t, A fixed. It asserts neither a uniform critical bubble tail nor a moving-template deformation inequality at an unspecified rate.

5 Finite Gram certificates and different discarded costs

Choose a finite observation region $L \subset V$. Write

$$H_L = \Pi_L H, \qquad H_Q = (I - \Pi_L)H, \qquad G_L = H_L^* H_L, \quad G_Q = H_Q^* H_Q.$$

The essential identity is

$$G_S = G_L + G_Q, \qquad G_Q \succeq 0. \tag{5}$$

There is no retained/discarded cross term: the two row spaces are orthogonal. A tail certificate is an upper bound on the actual infinite G_Q, not an inference from the size of the observed finite matrix.

Proposition 5.1 (Quadratic column-tail budget)**.** *Suppose certified numbers $\beta_s \geq 0$ satisfy $\sum_{x \notin L} T(x,s)^2 \leq \beta_s^2$. Put $\tau^2 = \sum_s \beta_s^2$. If a finite matrix $\widehat{H}_L$ satisfies $\left\|H_L - \widehat{H}_L\right\| \leq \varepsilon$, and $a_0 \geq \left\|\widehat{H}_L\right\|$ is certified, then*

$$\|\Delta\| \leq (a_0 + \varepsilon)^2 + \tau^2. \tag{6}$$

If instead a real symmetric matrix $E_Q \succeq G_Q$ is certified, one may use the sharper finite matrix bound

$$\|\Delta\| \leq \lambda_{\max}(\widehat{H}_L^* \widehat{H}_L + E_Q) + 2a_0\varepsilon + \varepsilon^2. \tag{7}$$

Proof. Cauchy–Schwarz bounds $|G_Q(s,r)| \leq \beta_s \beta_r$. Hence $\|G_Q\| \leq \tau^2$; equivalently $\|H_Q\|^2 \leq \operatorname{tr} G_Q \leq \tau^2$. Expanding $H_L = \widehat{H}_L + F$ bounds the Gram difference in norm by $2a_0\varepsilon + \varepsilon^2$. Now use (5) and Theorem 4.1. □

One must not infer $G_Q \preceq \beta\beta^*$ just from entrywise bounds: a rank-one matrix does not generally dominate a full-rank Gram matrix in Loewner order. The scalar estimate $G_Q \preceq \tau^2 I$ is always justified. A better E_Q needs a separate positive-semidefinite certificate. For example, partition the omitted vertices into shells Q_j and certify each shell Gram by $E_j \succeq H_{Q_j}^* H_{Q_j}$ with $\sum_j \|E_j\| < \infty$; then $E_Q = \sum_j E_j$ is legitimate.

Proposition 5.2 (Approximating the loss itself)**.** *Let $a \geq \|H_L\|$, $b \geq \|H_Q\|$. Then*

$$\|\Delta - \Pi_L \Delta \Pi_L\| \leq \Phi(a,b) := \frac{b^2 + \sqrt{b^4 + 4a^2b^2}}{2}. \tag{8}$$

In particular $\Phi(a,b) \to 0$ as $b \to 0$, for bounded a. An additional finite-section error ε_Δ adds linearly to (8).

Proof. The retained–retained block of the difference vanishes. Its two off-diagonal blocks have norms at most ab, by entrywise domination with $H_L H_Q^*$ and its transpose, and its discarded–discarded block has norm at most b^2. For $f = f_L + f_Q$, dominate the absolute quadratic form by

$$2ab\,\|f_L\|\,\|f_Q\| + b^2\,\|f_Q\|^2\,.$$

The largest eigenvalue of $\left[\begin{smallmatrix} 0 & ab \\ ab & b^2 \end{smallmatrix}\right]$ is $\Phi(a,b)$, giving the self-adjoint operator norm bound. The other eigenvalue has no larger absolute value. □

The distinction is material: (6) has a *quadratic* omitted-column cost b^2, whereas (8) is generally of order ab. Taking $H = (a,b)^T$ and $\Delta = HH^*$ makes (8) exact. It is therefore false to transfer the quadratic Gram cost to finite-section approximation of the loss operator itself.

5.1 Uniformity, geometry, and the source's moving corridor

Proposition 5.3 (Uniform local family)**.** *For a family of symmetric losses with interfaces S_θ, assume a uniform bound $\|H_\theta\| \leq M_H$. If finite regions $L_{\theta,R}$ satisfy*

$$\sup_\theta \sum_{s \in S_\theta} \sum_{x \notin L_{\theta,R}} T_\theta(x,s)^2 \leq \eta_R^2 \longrightarrow 0,$$

then

$$\sup_\theta \left\|\Delta_\theta - \Pi_{L_{\theta,R}} \Delta_\theta \Pi_{L_{\theta,R}}\right\| \leq \Phi(M_H, \eta_R) \longrightarrow 0.$$

Proof. Use $a \le M_H$, $b \le \eta_R$ in Proposition 5.2. The function Φ is increasing in its nonnegative arguments, as is also clear from its nonnegative 2-by-2 matrix representation. □

A uniform bound on the *total* bubble sums does not alone give the stated tail tightness: unit masses can move arbitrarily far outside every chosen region. In the source, a corridor has an interface contained in the closed neighborhood of its internal path, of size at most $(d+1)(k-1)$. As $k \to \infty$, both this size and the region needed for each tail must be accounted for. Quasi-transitivity helps reduce column *types*, but does not remove the factor for the number of columns or provide a uniform rate as $p \uparrow p_c$.

6 An exact regular-tree model

Let G be the $(q+1)$-regular tree, $q \ge 2$, and let $\mathcal{A}$ denote its adjacency operator. Unique paths give $T_p(x,y) = p^{\operatorname{dist}(x,y)}$. The following classical model is a useful test of the distinction between rows, columns, finite sections, and infinite-range operators. Its formulas are derived here, not presented as new percolation theory.

6.1 Susceptibility, bubbles, and the correct resolvent branch

The size of a distance-n sphere is $(q+1)q^{n-1}$ for $n \ge 1$. Therefore

$$\chi_p = \frac{1+p}{1-qp} \qquad (p < 1/q), \qquad \chi_p = \infty \qquad (p \ge 1/q), \tag{9}$$

$$b(p) = \sum_x T_p(o,x)^2 = \frac{1+p^2}{1-qp^2} \qquad (p < q^{-1/2}). \tag{10}$$

The branching-process threshold $p_c = 1/q$ is classical; the root has $q+1$ offspring trials and subsequent vertices have q, so extinction is determined by the latter mean qp. At criticality the nondegenerate critical offspring law becomes extinct almost surely, while its total expected progeny diverges.

The pointwise resolvent identity is

$$((1+qp^2)I - p\mathcal{A})\,T_p = (1-p^2)I. \tag{11}$$

For a vertex at positive distance n from the column root, one neighbor has distance $n-1$ and q have distance $n+1$; their terms cancel. At the root the remaining value is $1-p^2$. For $p < q^{-1/2}$, the column is in ℓ^2 by (10). Also $\|\mathcal{A}\| = 2\sqrt{q}$, so the left factor is invertible with lower spectral bound $(1-p\sqrt{q})^2 > 0$. Consequently its inverse has precisely these columns and

$$\|T_p\| = \frac{1-p^2}{(1-p\sqrt{q})^2}, \qquad 0 \le p < q^{-1/2}. \tag{12}$$

For completeness, the adjacency norm upper bound follows by the weighted Schur test using an end-oriented height b and the positive weight $h(x) = q^{-b(x)/2}$, with one neighbor at height $b-1$ and q at $b+1$. Then $\mathcal{A}h = 2\sqrt{q}\,h$. The finitely supported radial functions $f_N(x) = q^{-\operatorname{dist}(o,x)/2}\mathbf{1}_{\{\operatorname{dist}(o,x)\le N\}}$ have Rayleigh quotient tending to $2\sqrt{q}$, by summing the N edge layers. Thus the upper spectral edge is attained as a spectral limit, which gives equality in (12).

The formal identity (11) alone is not enough at $p > q^{-1/2}$. There the indicated columns are not in ℓ^2, even though the algebraic denominator is again nonzero. A bounded inverse on that branch does not have the connection columns. At $p = q^{-1/2}$ the denominator vanishes and the bubble sum diverges. This is a concrete warning against identifying a formal inverse with the physical connection kernel.

6.2 Exact finite-interface Gram matrix

If $\ell = \mathrm{dist}(s, r)$ and $p < q^{-1/2}$, then

$$G_S(s, r) = p^\ell \left(\ell + 1 + \frac{((q-1)\ell + q + 1)p^2}{1 - qp^2} \right). \tag{13}$$

Indeed project a summation vertex x onto the geodesic from s to r. The $\ell+1$ vertices on the geodesic contribute $(\ell+1)p^\ell$. For $\ell \geq 1$, the off-geodesic branches have $2q + (\ell-1)(q-1) = (q-1)\ell + q + 1$ initial edges and subsequent branching q. Distance $j \geq 1$ off the geodesic contributes a factor p^{2j}. The case $\ell = 0$ has $q+1$ initial branches and gives the same formula.

Take $L = \bigcup_{s \in S} B(s, R)$, $R \geq 0$. Each column has the certified tail

$$\beta_R^2 = \frac{q+1}{q} \frac{(qp^2)^{R+1}}{1 - qp^2}, \qquad \|G_Q\| \leq |S| \beta_R^2. \tag{14}$$

The region omitted from the union is contained in the complement of every individual ball, which explains the inequality for the union. These are valid certificates throughout $p \in [1/q, \bar{p}]$ for every fixed $\bar{p} < q^{-1/2}$, although every susceptibility row in that interval diverges. The local operator theorem only needs the deletion inequality where it is available; the Gram and tail formulas themselves hold throughout this interval.

For a numerical scale with exact constants, take $q = 2$, $p = 1/2$, and delete $A = \{o\}$. The required collar S has four vertices. Its diagonal Gram entries are $5/2$, root–neighbor entries are 2, and distinct-neighbor entries are $11/8$. Hence

$$\lambda_{\max}(G_S) = \frac{31 + \sqrt{889}}{8}, \qquad |S| \beta_R^2 = 6\, 2^{-R}.$$

The two neighbor-difference eigenvalues are $9/8$; the remaining two come from the root and the normalized constant vector on its three neighbors. Thus a tail-energy tolerance $\zeta > 0$ is met by any $R \geq \max(0, \lceil \log_2(6/\zeta) \rceil)$. This is a real discarded-tail certificate, not a finite-ball eigenvalue extrapolation.

6.3 The undeformed deletion is not positive semidefinite

At $t = 0$, deleting o splits the tree into $q+1$ branches. Let $h_i(x) = p^{\mathrm{dist}(o,x)}$ on branch i and zero elsewhere, and set

$$v = \|h_i\|^2 = \frac{p^2}{1 - qp^2}.$$

In the orthonormal basis $e_o, h_1/\sqrt{v}, \ldots, h_{q+1}/\sqrt{v}$, the nonzero part of the deletion operator for $p > 0$ is

$$\begin{pmatrix} 1 & \sqrt{v}\, \mathbf{1}^T \\ \sqrt{v}\, \mathbf{1} & v(\mathbf{1}\mathbf{1}^T - I) \end{pmatrix}. \tag{15}$$

This follows by distinguishing the same branch, different branches, and a terminal equal to o. It has eigenvalue $-v$ on each branch-difference vector, and two remaining eigenvalues

$$\lambda_\pm = \frac{1 + qv \pm \sqrt{(1-qv)^2 + 4(q+1)v}}{2}.$$

Its norm is λ_+, and its squared Hilbert–Schmidt norm is $1 + 2(q+1)v + q(q+1)v^2$. For $q = 2, p = 1/2$, the exact norm is $1 + \sqrt{3/2}$, substantially below the general collar Gram upper bound. The negative eigenvalues are a direct percolation example, not just an arbitrary matrix counterexample to presumed positivity.

7 Whole-operator certificates are a different task

Proposition 7.1 (Finite-coordinate obstruction)**.** *If V is infinite and T_p is bounded, then for every finite $L \subset V$,*

$$\|T_p - \Pi_L T_p \Pi_L\| \geq 1.$$

In fact the essential norm of T_p is at least one.

Proof. For $x \notin L$, the matrix coefficient at e_x, e_x of the difference is $T_p(x,x) = 1$, giving the first statement. For a compact operator C, along any sequence of distinct vertices x_n, $\|Ce_{x_n}\| \to 0$. Therefore $\|T_p - C\| \geq \limsup_n |\langle e_{x_n}, (T_p - C)e_{x_n}\rangle| = 1$. □

At $p = 0$, $T_p = I$, making the obstruction immediate. Finite sections do give lower bounds $\|\Pi_L T_p \Pi_L\| \leq \|T_p\|$, and exhaustion recovers the norm as a supremum when boundedness is known or the supremum is finite. A list of finitely many such lower bounds cannot certify the needed upper bound.

7.1 Weighted Schur residuals with a genuine infinite tail

Lemma 7.2 (Two-sided weighted Schur test)**.** *Let $E(x,y) \geq 0$, and suppose a strictly positive weight h satisfies, at every vertex,*

$$\sum_y E(x,y)h(y) \leq r h(x), \qquad \sum_x E(x,y)h(x) \leq c h(y).$$

Then any kernel F with $|F(x,y)| \leq E(x,y)$ extends boundedly on counting ℓ^2, with $\|F\| \leq \sqrt{rc}$. The weight need not be in ℓ^2 or bounded away from zero.

Proof. For finitely supported f, Cauchy–Schwarz gives

$$|Ff(x)|^2 \leq \left(\sum_y E(x,y)h(y)\right)\left(\sum_y E(x,y)\frac{|f(y)|^2}{h(y)}\right).$$

Sum over x, use both hypotheses and Tonelli, then density. □

If U is a bounded *infinite-rank finite-range* kernel and $|T_p - U| \leq E$ has this certificate, then $\|T_p - U\| \leq r_0 := \sqrt{rc}$. It is permissible to reduce all these sums to finitely many types only if the reduction, its symmetry, and the omitted shells are proved for every vertex. Finite-range does not mean finite-rank. A polynomial in a graph-supported operator has finite propagation but acts on the whole graph.

The regular tree demonstrates such a certificate constructively. Put

$$U_R(x,y) = p^{\mathrm{dist}(x,y)}\mathbf{1}_{\{\mathrm{dist}(x,y)\leq R\}}, \qquad x_0 = p\sqrt{q} < 1.$$

For the height weight used above, the sphere operator S_n satisfies, for $n \geq 1$,

$$S_n h = q^{n/2}\frac{q+1+(q-1)n}{q}\,h.$$

This follows from $S_1 = \mathcal{A}$, $S_2 = \mathcal{A}^2 - (q+1)I$, and $S_{n+1} = \mathcal{A}S_n - qS_{n-1}$ for $n \geq 2$, or from the parent/child decomposition of paths. Thus both weighted Schur sums of $T_p - U_R$ equal the weight times

$$\mathcal{E}_R = \frac{x_0^{R+1}}{q}\left[\frac{q+1+(q-1)(R+1)}{1-x_0} + \frac{(q-1)x_0}{(1-x_0)^2}\right]. \tag{16}$$

Consequently $\|T_p - U_R\| \le \mathcal{E}_R \to 0$. Every U_R here is infinite rank; its diagonal is one. The unweighted row-tail sum is infinite when $p \ge 1/q$, so replacing this weight by $h = 1$ would destroy the certificate. The height weight is used only in the proof of a norm on counting ℓ^2; no hidden change of Hilbert measure occurs.

8 Sprinkling and deformation residual interfaces

Let $\mathcal{A}$ now denote adjacency on the original bounded-degree graph. Fix p for which $T = T_p$ is bounded, $M \ge \|T\|$, and let $0 \le a < 1$. The source's BK decomposition of a path into old connections and newly opened edges gives the entrywise inequality

$$T_{p+(1-p)a} \le \sum_{j=0}^{\infty} a^j T(\mathcal{A}T)^j. \tag{17}$$

This path-counting input is retained from the source. The following quantification is the classical Neumann estimate applied in the correct infinite space and factor order.

Proposition 8.1 (Certified sprinkling width and tail)**.** *Suppose $b \ge \|\mathcal{A}T\|$ is an actual whole-operator bound and $ab < 1$. Define*

$$R_a = T(I - a\mathcal{A}T)^{-1}, \qquad R_{a,J} = \sum_{j=0}^{J} a^j T(\mathcal{A}T)^j.$$

Then

$$\left\|T_{p+(1-p)a}\right\| \le \frac{M}{1-ab}, \qquad \|R_a - R_{a,J}\| \le \frac{M(ab)^{J+1}}{1-ab}. \tag{18}$$

If $p = p_c$ and $M < \infty$ has independently been proved, every $a \in (0, \min(1, 1/b))$ gives a certified point above p_c with bounded two-point operator.

Proof. Submultiplicativity gives absolute operator-norm convergence. Both products with $I - a\mathcal{A}T$ telescope, with remainder tending to zero. The geometric series gives the tail estimate. Each term has a nonnegative kernel, so the limit kernel majorizes the left side of (17); entrywise domination bounds its norm by $\|R_a\|$. □

The finite polynomial $R_{a,J}$ is *not* by itself an upper bound on $T_{p+(1-p)a}$. It approximates the majorizing resolvent; the omitted infinite sum still needs (18). The number b may be smaller than dM, but computing a small finite-section norm of $\mathcal{A}T$ does not certify it. Nonnormality is also relevant: spectral radius alone does not give the stated geometric tail with unit prefactor.

Corollary 8.2 (A usable error budget)**.** *Suppose $\|T - U\| \le r_0$ is certified on the whole graph, $\alpha \ge \|\mathcal{A}\|$, $b_U \ge \|\mathcal{A}U\|$, and $M_U \ge \|U\|$. Then in Proposition 8.1 one may use*

$$b = b_U + \alpha r_0, \qquad M = M_U + r_0.$$

The sufficient width condition is $a(b_U + \alpha r_0) < 1$.

Proof. Use $\mathcal{A}T = \mathcal{A}U + \mathcal{A}(T - U)$ and the triangle inequality. All estimates concern bounded operators on the same Hilbert space. □

Even the simpler choice $b = \|\mathcal{A}\|\, M$ can improve the source's maximum-degree bound dM. This is a classical local certificate improvement, not a new phase theorem. Block operator norm majorants are also permissible on an orthogonal decomposition if they dominate *all* blocks: Cauchy–Schwarz bounds $\|Bf\|$ by the nonnegative scalar block matrix acting on $(\|f_i\|)_i$. Discarded blocks remain obligations; a finite block truncation is not enough.

8.1 Keeping the source's sharper sandwich

The source already proves

$$\left\|T - K_t^V\right\| \le 2t \left\|TP^kT\right\| \le 2t\rho^k \|T\|^2 . \tag{19}$$

Retaining the middle expression is therefore not a new result. Its certified transport to an approximate whole-operator model is: if $\|T - U\| \le r_0$, $\|U\| \le M_U$, and $C_k \ge \left\|UP^kU\right\|$, then

$$\left\|TP^kT\right\| \le C_k + \rho^k r_0(2M_U + r_0). \tag{20}$$

Indeed, expand $T = U + F$, and bound the two cross terms and the quadratic term. Neither T and P nor U and P need commute. Combining (19)–(20) gives a sufficient norm-survival condition

$$2t(C_k + \rho^k r_0(2M_U + r_0)) \le \vartheta m_{\text{low}}, \qquad \|T\| \ge m_{\text{low}} > 0,$$

which implies $\left\|K_t^V\right\| \ge (1 - \vartheta)m_{\text{low}}$ for $0 \le \vartheta < 1$. A finite section may supply m_{low}, but cannot supply the upper bounds C_k, r_0, M_U without whole-graph control. No positivity of K_t^V is invoked.

9 Original Ram proofs and the finite/infinite boundary

The original private `TR-core-pat-5` [6] explicitly types its finite graph operators and metrics, distinguishes directed transitions from symmetric Laplacians, proves a norm-controlled Neumann tail with the factor order retained, and warns that a spectral-radius condition alone does not give a prescribed-norm tail estimate. Its retained/discarded proposition is a Pythagorean identity with a separate complementary error. Its Schur residual identity transports an interior defect into the boundary rows; small reduced residual is not a full solution certificate. These actual proofs, not just a catalogue description, were read freshly for this report.

We use those principles in three precisely delimited ways. First, the orthogonality in (5) explains why discarded *column energy* enters quadratically, while (8) retains the off-diagonal effect on the physical loss. Second, the finite Neumann proof extends to a Banach operator algebra only after a true infinite-space contraction is supplied; Proposition 8.1 provides that extension. Third, a reduced numerical model must document the operator type and coordinate metric. Orbit-root averaging is not a replacement for counting-measure $\ell^2(V)$.

The freshly read original Volume 21 chapter on spectral limits [7] gives an explicit diagonal self-adjoint operator with spectrum $\mathbb{Z} + 1/4$, together with finite-dimensional compressions having a false zero mode. A zero projected residual can coexist with a large full residual. Its two-sided, domain-aware inverse defects and approximate-dual residual identity explain why a finite physical matrix cannot simply be promoted to an infinite uniform inverse. In our bounded setting the much simpler diagonal-one obstruction in Proposition 7.1 suffices. Neither private original proves the source's percolation amplification or a global threshold gap. These are local operator methods with explicitly transported assumptions, not a claim that earlier finite mathematics settles the conjecture.

10 Advantages, costs, and limits

The main gain is not a faster global percolation solver. It is an alternative certified question that remains finite-interface when row sums become infinite:

finite deletion $\longrightarrow$ finitely many connection columns $\longrightarrow$ bubble Gram and a compact loss.

The theorem needs no unimodularity, invariant trace, or global operator norm once the deletion inequality and the finite column hypotheses are supplied. Its endpoint form can therefore be useful independently of the final source contradiction. The Gram certificate retains correlations between interface columns; replacing it by $\sum_s b_s$ is safe but often wasteful. The regular-tree example quantifies both that gain and the remaining slack.

The costs are equally concrete. Connection probabilities in the finite region may themselves be hard to certify. The exterior energy bounds remain infinite obligations. Increasing the interface can increase dimension, conditioning, and tail cost. Hilbert–Schmidt compactness is not a finite-rank formula for the actual loss. Approximation of that loss generally costs the first order of the omitted-column norm, even though its global norm upper bound uses squared tail energy. The square-summability premise is not deduced here for all critical graphs in the assigned class.

No mathematical error in the source is asserted by this report. The source already states the sharper sandwich, avoids positivity of the deformation, retains the blocker collar, and supplies careful non-independent exploration arguments. We preserve these features. Our local additions do not replace its all-order critical moment argument, joint splicing inverse, triangle differential inequality, escape-path construction, or the simultaneous-coupling phase argument. They do not constitute an independent verification of $p_c < p_{2\to 2}$.

Promising further work is to produce uniform shell-Gram envelopes for graph families with controlled geometry, to optimize the finite observation regions against correlated columns, and to track the interface size in a moving k, p deformation. For the latter, one would need an explicit common tail modulus and all of the source's probabilistic constants in the same parameter regime. A compactness theorem for each fixed interface is not a uniform theorem along that sequence. Any claimed computational gain should also include the cost of obtaining the probabilities and exterior certificates, not merely the dimension of the last Gram matrix.

11 Reproducible evidence and contribution assessment

The public read ledger covers all 17 cached text files: 4,421 literal lines, including 4,045 lines in 15 TeX files, 356 bibliography lines, and 20 README lines. Private original reading covers 183 lines of `TR-core-pat-5` (70–133, 189–248, 287–309, 364–399) and the full 242-line Volume 21 spectral-limits chapter. The uninspected complement of the first original and the rest of the private corpus remain explicitly unread for this assignment. An external figure referenced by that chapter was not visually inspected. Pre-read and post-read SHA-256 values are recorded per file. The parent snapshot authentication is provenance only; its checks and read intervals are not counted as author work.

The independently authored exact script `M386-checks.py` passes 1,899 parameter fixtures and 8,424 assertions:

Fixture family	Cases
Finite Gram, Hilbert–Schmidt, and bilinear inequalities	500
Sharp cross-block characteristic identity	81
Enumerated tree pairs plus exact exterior sums	924
Tree column-tail identity	36
Tree deletion spectrum identities	21
Infinite-range weighted Schur tail identity	252
Ordered Neumann identity and finite norm tail	84
Printed star-Gram constants	1

All acceptance arithmetic is rational. The finite Neumann fixtures use an explicitly stated ℓ^∞ norm and check factor order and geometric remainders; they are not numerical tests of an infinite ℓ^2 threshold. The general operator inequalities are justified by the written proofs, not by sampling. No foreign checker was executed.

Applicability stays L3. C3 is assigned for the source-specific compact deletion extension, endpoint interface, and the distinction between quadratic Gram-tail certification and sharp cross-block approximation cost. Schur's test, Neumann series, Gram factorization, regular-tree spectral facts, and finite exact arithmetic receive no invented novelty credit. No C4 claim is made: there is no new global phase theorem, uniform critical tail modulus for the whole graph class, or demonstrated runtime improvement.

References

[1] OpenAI Math repository, *Nonuniqueness of percolation on nonamenable quasi-transitive graphs*, September 24, 2026. Assigned cached source, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. https://github.com/openai/math/tree/main/preprints/Nonuniqueness-of-percolation-on-nonamenable-quasi-transitive-graphs-September-24-2026. Live URL is a public locator, not the immutable identity.

[2] T. Hutchcroft, *The L^2 boundedness condition in nonamenable percolation*, Electronic Journal of Probability 25 (2020), paper 127. https://arxiv.org/html/1904.05804v2. Introduction and Remark 2.3 inspected; external proof not independently audited in full.

[3] T. Hutchcroft, *Percolation on hyperbolic graphs*, Geometric and Functional Analysis 29 (2019), 766–810. https://arxiv.org/abs/1804.10191. Primary abstract and source bibliography used for scope.

[4] I. Choi and D. Seo, *Percolation in acylindrically hyperbolic groups*, arXiv:2508.08932v2, August 13, 2025. https://arxiv.org/abs/2508.08932. Fresh abstract/metadata reading only.

[5] T. Hutchcroft and M. Pan, *Dimension jump at the uniqueness threshold for percolation in $\infty + d$ dimensions*, arXiv:2412.15895v1, December 20, 2024. https://arxiv.org/html/2412.15895v1. Theorem 1.2 statement inspected; not a full-proof audit.

[6] M. Subbarao, original private `TR-core-pat-5`, typed graph operators, norm-controlled inverse, retained/discarded error, and Schur residual proofs. Local confidential corpus; literal intervals specified in Section 11.

[7] M. Subbarao, original private Ram Calculus book, Volume 21, `ch01f-spectral-limits.tex`, 242 lines. Spectral pollution, domain-aware inverse, and target residual proofs. Local confidential corpus; no private text uploaded.